

SKILLS

Programming Languages	Python, SQL, HTML, CSS, Git, PowerBI
Frameworks & Libraries	Flask, Scikit-learn, XGBoost, Hugging Face Transformers
Data Science & Machine Learning	Scikit-learn, TensorFlow, Natural Language Processing (NLP), Data Visualization, Generative AI , Statistical Modeling & Analysis, AI Agents
Cloud Technologies	AWS, CloudFormation, Amazon EC2, Amazon S3, AWS Lambda, AWS IAM, Docker, Terraform
Web Development & API	FastAPI, Backend Development , React
Quantitative Skills	Econometrics, Microeconomics, Macroeconomics, VAR, Risk Modeling, Credit Risk Forecasting

EDUCATION

M.S. Data Science & AI, University of Central Missouri	Jan 2025 — Dec 2026 (expected)
Master of Business Administration (MBA), Asia e University	Sep 2023 — Aug 2025
B.E. Computer Engineering, Tribhuvan University	Nov 2017 — Aug 2022

TECHNICAL EXPERIENCE

Graduate Research Assistant University of Central Missouri	Jan 2025 — Present Lee's Summit, MO
- Build and benchmark retrieval-augmented generation (RAG) pipelines for biomedical question answering combining BM25/embedding retrieval, reranking, and summarization across factoid, list, and yes/no tasks with ablation studies across chunking, embedding models, and prompting strategies to improve answer faithfulness and relevance. - Develop recommendation-system baselines with standardized metric reporting (HR@K, NDCG@K, Recall@K) for reproducible model comparison. - Develop an agentic AI system for cardiac care that fuses ECG waveform analysis with clinical record data, building the pipeline end to end signal preprocessing, arrhythmia classification, signal-quality assessment, and a fusion layer where specialized agents reason over calibrated model outputs rather than raw predictions.	
Financial Data Analyst Genese Solution Pvt. Ltd.	Dec 2022 — Nov 2024 Nepal
- Built a natural-language-to-AWS deployment tool using an Anthropic LLM and CloudFormation, improving resource-creation efficiency by 20%. - Developed credit-risk forecasting models in Python (Scikit-learn, TensorFlow), improving accuracy by 15% and reducing loan defaults for banking partners. - Standardized code review and testing (90% fewer deployment defects); automated payroll processing with SQL and Python, saving 15 hours weekly.	
Financial Data Analyst Apprentice Code Rush	Aug 2022 — Nov 2022 Nepal
- Built stock-price forecasting models with VAR and time series analysis, improving accuracy by 15%. - Assessed portfolio risk using Monte Carlo simulation, Value at Risk (VaR), and Expected Shortfall at 95% confidence.	

PROJECTS

Generative-AI AWS Provisioning Platform	2024
Led backend development of a multi-tenant platform that provisions AWS resources from natural-language prompts via LLM prompt engineering; built 10+ FastAPI endpoints and raised provisioning throughput by 20%.	
Algorithmic Trading & Synthetic Index Modeling (Kalshi Markets)	2025
Built a high-frequency trading system in Rust with a Deep Reinforcement Learning agent (DQN, PyTorch) for arbitrage in crypto prediction markets; backtested strategies and deployed a fault-tolerant service with ONNX and a real-time Streamlit dashboard.	

PUBLICATIONS

- Stress-Testing Biomedical Retrieval-Augmented Generation: A Diagnostic Evaluation under Noise, Conflicting Evidence, and Unanswerability** — *IEEE International Conference on E-health Networking, Application & Services*, Oct 19, 2026.
- CodeBERT at the Watering Hole: Dual-Layer URL & JavaScript Attack Detection** — *2026 IEEE International Conference on Electro Information Technology (eIT)*, May 21, 2026.
- What Hurts RAG in Biomedical QA: A Stress-Test Study of Noise, Conflict & Unanswerability** — *MMLS 2026 (Lighting Talk and Poster)*.

CERTIFICATIONS

- AWS Certified AI Practitioner** — Amazon Web Services Training and Certification Expires Nov 2028